

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396893
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Jenkins; John et al.

Apparatuses, systems, and methods for increasing or manipulating noise attenuation in hearing protection device

Abstract

A protection cup for a hearing protection device (HPD) is provided. The protection cup includes a plurality of fractal elements disposed on an inner surface of the protection cup.

Inventors: Jenkins; John (Morris Plains, NJ), Hutt; Scott (Morris Plains, NJ), Sun; Wei (Shanghai, CN), Chen; Yongjun (Shanghai, CN), Jiang; Lei (Shanghai, CN)

Applicant: Honeywell Safety Products USA, Inc. (Charlotte, NC)

Family ID: 1000008780000

Appl. No.: 18/732425

Filed: June 03, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240315879 A1	Sep. 26, 2024

Related U.S. Application Data

continuation parent-doc US 17309879 US 12029628 WO PCT/CN2019/075562 20190220 child-doc US 18732425
us-provisional-application US 62786810 20181231

Publication Classification

Int. Cl.: A61F11/14 (20060101); B29D99/00 (20100101); G10K11/16 (20060101); G10K11/162 (20060101)

U.S. Cl.:

CPC A61F11/14 (20130101); B29D99/00 (20130101); G10K11/162 (20130101);

Field of Classification Search

CPC: A61F (11/14); A61F (11/06); B29D (99/00); G10K (11/162); G10K (11/16); E04B (2001/8428); E04B (2001/8423); E04B (1/84)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2325150	12/1942	Sahlmann	N/A	N/A
2582907	12/1951	Kaufmann	N/A	N/A

2883671	12/1958	Hornickel	N/A	N/A
2981958	12/1960	Wadsworth et al.	N/A	N/A
3051961	12/1961	Clark	N/A	N/A
3197785	12/1964	Simpson et al.	N/A	N/A
3432861	12/1968	Flagg	N/A	N/A
3875592	12/1974	Aileo	N/A	N/A
D239787	12/1975	Lonnstedt	N/A	N/A
D244213	12/1976	Besasie	N/A	N/A
D276665	12/1983	Higginson	N/A	N/A
D276855	12/1983	Falco	N/A	N/A
D298670	12/1987	Palmaer	N/A	N/A
D301145	12/1988	Besasie et al.	N/A	N/A
D313092	12/1989	Nilsson	N/A	N/A
5068923	12/1990	Sjoqvist	N/A	N/A
D375584	12/1995	Westerdal	N/A	N/A
D385665	12/1996	Westerdal	N/A	N/A
5815842	12/1997	Hiselius	N/A	N/A
5996123	12/1998	Leight et al.	N/A	N/A
D427382	12/1999	Leight et al.	N/A	N/A
6353938	12/2001	Young	N/A	N/A
D459033	12/2001	Dix et al.	N/A	N/A
D459034	12/2001	Dix et al.	N/A	N/A
6449806	12/2001	Engelhard et al.	N/A	N/A
D465616	12/2001	Nilsson	N/A	N/A
D541483	12/2006	Birgersson et al.	N/A	N/A
7198133	12/2006	Warring et al.	N/A	N/A
D544650	12/2006	Westerdal	N/A	N/A
7308965	12/2006	Sapoval	181/293	E01F 8/0076
7444687	12/2007	Sato et al.	N/A	N/A
7604094	12/2008	Magyari	181/294	G10K 11/20
7717226	12/2009	Purcell et al.	N/A	N/A
D641725	12/2010	Chong et al.	N/A	N/A
D649956	12/2010	Gresko et al.	N/A	N/A
D651586	12/2011	Lee et al.	N/A	N/A
D651764	12/2011	Gresko et al.	N/A	N/A
D663900	12/2011	Karlsson et al.	N/A	N/A
D687189	12/2012	Fairclough et al.	N/A	N/A
D735419	12/2014	Rose et al.	N/A	N/A
D747557	12/2015	Bonney et al.	N/A	N/A
D752816	12/2015	Fletcher et al.	N/A	N/A
D753884	12/2015	Bonney et al.	N/A	N/A
D775605	12/2016	Lee et al.	N/A	N/A
D776081	12/2016	Willman	N/A	N/A
D787469	12/2016	Djohar et al.	N/A	N/A
D787470	12/2016	Kelly et al.	N/A	N/A
9640166	12/2016	Simon et al.	N/A	N/A
D789615	12/2016	Pratley	N/A	N/A
9744078	12/2016	Carolan et al.	N/A	N/A
D796477	12/2016	Minn et al.	N/A	N/A
9786261	12/2016	Jenkins	N/A	N/A
D805697	12/2016	Pratley	N/A	N/A
D807584	12/2017	Fletcher et al.	N/A	N/A
D815615	12/2017	Wernblad	N/A	N/A
D821653	12/2017	Pratley	N/A	N/A
10080077	12/2017	Silvestri	N/A	N/A
D830331	12/2017	Petersen et al.	N/A	N/A
D848396	12/2018	Wilson et al.	N/A	N/A
D848679	12/2018	Wilson et al.	N/A	N/A
D851839	12/2018	Scanlon	N/A	N/A
D856594	12/2018	Wilson et al.	N/A	N/A
D863690	12/2018	Chen	N/A	N/A
D864900	12/2018	Miller et al.	N/A	N/A
D874065	12/2019	Bui	N/A	N/A

D881475	12/2019	Xu	N/A	N/A
D885683	12/2019	Qiu	N/A	N/A
D886802	12/2019	Willis et al.	N/A	N/A
D891397	12/2019	Zeng	N/A	N/A
D913598	12/2020	Jesper et al.	N/A	N/A
11026843	12/2020	Werner et al.	N/A	N/A
D924200	12/2020	Ueno et al.	N/A	N/A
D925490	12/2020	Ueno et al.	N/A	N/A
D939779	12/2020	Jenkins et al.	N/A	N/A
D942089	12/2021	Jenkins et al.	N/A	N/A
D945981	12/2021	Rois	N/A	N/A
11369521	12/2021	Almeflo	N/A	N/A
D960459	12/2021	Jenkins et al.	N/A	N/A
11557271	12/2022	Su et al.	N/A	N/A
D978444	12/2022	Chen	N/A	N/A
11582548	12/2022	Bui	N/A	A61B 5/0006
D1013282	12/2023	Bos	N/A	N/A
D1033751	12/2023	Cooper et al.	N/A	N/A
12029628	12/2023	Jenkins	N/A	G10K 11/162
2007/0157365	12/2006	Hansson	N/A	N/A
2008/0075314	12/2007	Chang	N/A	N/A
2010/0158301	12/2009	Kuhtz et al.	N/A	N/A
2013/0340769	12/2012	Roos et al.	N/A	N/A
2019/0069065	12/2018	Brace	N/A	N/A
2019/0110930	12/2018	Hkansson	N/A	N/A
2020/0396532	12/2019	Bui et al.	N/A	N/A
2022/0062056	12/2021	Jenkins et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
203393862	12/2013	CN	N/A
105662708	12/2015	CN	N/A
106205590	12/2015	CN	N/A
106328115	12/2016	CN	N/A
106375896	12/2016	CN	N/A
109069299	12/2017	CN	N/A
110537219	12/2022	CN	B64D 29/00
3228290	12/2016	EP	N/A
2943000	12/2009	FR	B60R 13/0884
94/17763	12/1993	WO	N/A
2018/150351	12/2017	WO	N/A

OTHER PUBLICATIONS

Lucid Audio, HEARMUFFS™ Sounds, [retrieved Apr. 4, 2019] retrieved from the Internet <URL:

<https://lucidaudio.com/product/hearmuffs-sounds/>, 16 pages. cited by applicant

Lucid Audio, Kids HearMuffs™ Trio, [retrieved Apr. 4, 2019] retrieved from the Internet <URL:

<https://lucidaudio.com/product/kids-hearmuffs-trio/>, 14 pages. cited by applicant

Sapoval, B., et al., “Acoustical properties of irregular and fractal cavities”, Oct. 1, 1997, J. Acoust. Soc. Am., pp. 2014-2019, vol. 102, No. 4. cited by applicant

26 Max NRR, Hard Hat Cap Mounted Non-Conductive Earmuffs, <https://www.mscdirect.com/browse/tnpla/46721411>, 2022. (Year: 2022). cited by applicant

3M Twin-Cup Industrial Hearing Protector, <https://gemplers.com/products/3m-twin-cup-industrial-hearing-protector>, May 1, 2020. (Year: 2020). cited by applicant

CA Office Action Mailed on Jun. 7, 2023 for CA Application No. 3124461, 4 page(s). cited by applicant

CA Office Action Mailed on Oct. 6, 2022 for CA Application No. 3124461, 3 pages. cited by applicant

Corrected Notice of Allowability received for U.S. Appl. No. 29/680,786, mailed on Nov. 19, 2021, 5 pages. cited by applicant

Corrected Notice of Allowability received for U.S. Appl. No. 29/680,788, mailed on Nov. 19, 2021, 5 pages. cited by applicant

Ex Parte Quayle Action received for U.S. Appl. No. 29/680,786, mailed on Jul. 15, 2021, 15 pages. cited by applicant

Ex Parte Quayle Action received for U.S. Appl. No. 29/680,788, mailed on Jul. 15, 2021, 9 pages. cited by applicant

Examination Report issued in European Community Design Application No. 006738217 on Aug. 26, 2019, 5 pages. cited by applicant

Examination Report issued in European Community Design Application No. 006740239 issued on Aug. 27, 2019, 2 pages. cited by applicant

Extended European Search Report dated Sep. 6, 2022 for EP Application No. 19907300.8, 7 pages. cited by applicant
International Search Report and Written Opinion issued in International Application No. PCT/CN2019/075562 on Sep. 2, 2019, 7 pages. cited by applicant
Jackson H70 Vibe (Registered) Super Premium Dielectric Ear Muffs | Cap-Mount | NRR 25DB, https://cleanflow.net/products/jackson-super-prem-dielectric-ear-muffs-cap-mount-nrr-25db?currency=CAD&variant=39284694483005&utm_medium=cpc&utm_source=google&utm_campaign=Google%20Shopping&srsltid=AOLKFij3KOAFZHSnK6W3P76tL6463k064
copyright 2022 CleanFlow. (Year: 2022). cited by applicant
Non-Final Rejection Mailed on Sep. 21, 2023 for U.S. Appl. No. 17/309,879, 7 page(s). cited by applicant
Notice of Allowance and Fees Due (PTOL-85) Mailed on Mar. 20, 2024 for U.S. Appl. No. 17/309,879, 7 page(s). cited by applicant
Notice of Allowance received for U.S. Appl. No. 29/680,786, mailed on Oct. 5, 2021, 8 pages. cited by applicant
Notice of Allowance received for U.S. Appl. No. 29/680,788, mailed on Oct. 6, 2021, 8 pages. cited by applicant
Notice of Allowance received for U.S. Appl. No. 29/816,393, mailed on Apr. 4, 2022, 12 pages. cited by applicant
Notice of Allowance received for U.S. Appl. No. 29/816,393, mailed on Jun. 9, 2022, 6 pages. cited by applicant
Notice of Grant of Patent Right for a Design issued in Chinese Application No. 201930453176.6 on Jan. 6, 2020, 4 pages. cited by applicant
Intention to grant Mailed on Jan. 3, 2025 for EP Application No. 19907300, 9 page(s). cited by applicant
Honeywell Howard Leight VeriShield VS130HHV Hard Hat, Hi-Viz Earmuff (NRR 27), <https://www.earplugstore.com/howard-leight-verishield-vs130hhv-earmuff.html>, Copyright 2025 Ear Plug Superstore, retrieved from the Internet on Mar. 11, 2025, 5 pages. cited by applicant
Honeywell VeriShield Safety Earmuffs Ear Protection, Defender, NRR 35 dB Ear Muff (Pack of 1), https://www.flipkart.com/honeywell-verishield-safety-earmuffs-ear-protection-defender-nrr-35-db-muff/p/itm71ac36f9950f4?pid=EARGETABZAYXGHAX&cmpid=product.share.pp&_refld=pp.959aa35c-7f9b, Apr. 2023, retrieved from the Internet on Mar. 11, 2025, 3 pages. cited by applicant
Non-Final Rejection Mailed on Mar. 18, 2025 for U.S. Appl. No. 29/844,484, 11 page(s). cited by applicant

Primary Examiner: Martin; Edgardo San

Attorney, Agent or Firm: Fredrikson & Byron, P.A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/309,879, filed Jun. 25, 2021, which is a National Stage Entry of PCT Application No. PCT/CN2019/075562, filed Feb. 20, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/786,810, filed Dec. 31, 2018, the entire contents of each are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present disclosure relates generally to apparatuses, systems, and methods for hearing protection devices, and more particularly, to apparatuses, systems, and methods for improving and/or manipulating noise attenuation of hearing protection devices.

BACKGROUND

(2) A hearing protection device (also known as HPD) refers to a protection device that, for example, may be worn in or over the ears to reduce the level of noise entering the ear. HPDs are used in many situations. For example, when a user is exposed to hazardous noise, he may prefer to wear a HPD to prevent noise-induced hearing loss. HPDs may include, for example, earmuffs, earplugs, and other devices that may block or reduce the passage of environmental sound to a user's ear canal.

(3) Existing devices are plagued by many limitations and restrictions. For example, they do not provide sufficient noise attenuation. In addition, they are not tuned to any particular frequency of environmental noise.

BRIEF SUMMARY

(4) Various embodiments described herein relate to methods, apparatuses, and systems for improving the performance of hearing protection devices. In particular, various embodiments increase noise attenuation by disposing a plurality of fractal elements on an inner surface of a protection cup for a hearing protection device.

(5) In accordance with various embodiments, a protection cup for a hearing protection device is provided. The protection cup may comprise an outer surface, an inner surface on an opposite side from the outer surface, and a plurality of fractal elements disposed on the inner surface. The plurality of fractal elements may comprise a plurality of first fractal elements protruding from the inner surface of the protection cup, and a plurality of second fractal elements protruding from the inner surface of the protection cup. The plurality of first fractal elements and the plurality of second fractal elements form a plurality of regions on the inner surface.

(6) In some embodiments, the plurality of first fractal elements are longitudinal fractal elements, and the plurality of second fractal elements are transverse fractal elements. The transverse fractal elements are positioned perpendicular to the longitudinal fractal elements.

(7) In some embodiments, the plurality of fractal elements further comprise a plurality of additional fractal elements, and the plurality of additional fractal elements are disposed within at least one of the plurality of regions on the inner surface.

- (8) In some embodiments, each of the plurality of additional fractal elements comprises a longitudinal segment and at least one transverse segment.
- (9) In some embodiments, two or more of the plurality of additional fractal elements are connected through longitudinal segments of the two or more of the plurality of additional fractal elements.
- (10) In some embodiments, each of the plurality of additional fractal elements further comprises a plurality of leg segments, wherein the plurality of leg segments are disposed at end points of the at least one transverse segment.
- (11) In some embodiments, the plurality of fractal elements are molded on the inner surface of the protection cup.
- (12) In some embodiments, the plurality of additional fractal elements are of the same size.
- (13) In some embodiments, the inner surface of the protection cup comprises a protruding portion.
- (14) In some embodiments, the plurality of fractal elements further comprise a plurality of diagonal fractal elements.
- (15) In accordance with various embodiments, a method for manufacturing a protection cup for a hearing protection device is provided. The method comprises molding an outer surface, molding an inner surface on an opposite side from the outer surface, and molding a plurality of fractal elements on the inner surface. Molding a plurality of fractal elements on the inner surface further comprises molding a plurality of first fractal elements protruding from the inner surface of the protection cup, molding a plurality of second fractal elements protruding from the inner surface of the protection cup. The plurality of first fractal elements and the plurality of second fractal elements form a plurality of regions on the inner surface. In some embodiments molding the plurality of fractal elements on the inner surface further comprises adjusting the plurality of fractal elements based on attenuation ratings.
- (16) The foregoing illustrative summary, as well as other exemplary objectives and/or advantages of the disclosure, and the manner in which the same are accomplished, are further explained in the following detailed description and its accompanying drawings.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The description of the illustrative embodiments may be read in conjunction with the accompanying figures. It will be appreciated that, for simplicity and clarity of illustration, elements illustrated in the figures have not necessarily been drawn to scale, unless described otherwise. For example, the dimensions of some of the elements may be exaggerated relative to other elements, unless described otherwise. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the figures presented herein, in which:
- (2) FIG. 1A illustrates a left view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (3) FIG. 1B illustrates a right view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (4) FIG. 1C illustrates another view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (5) FIG. 1D illustrates a cross-sectional view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (6) FIG. 1E illustrates a back view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (7) FIG. 1F illustrates a front view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (8) FIG. 1G illustrates a top view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (9) FIG. 1H illustrates a bottom view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (10) FIG. 1I illustrates an isometric view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (11) FIG. 2 illustrates example measurements of an example fractal element in accordance with various embodiments of the present disclosure;
- (12) FIGS. 3A and 3B illustrate cross-sectional views of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (13) FIGS. 4A and 4B illustrate cross-sectional views of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (14) FIG. 5 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (15) FIG. 6 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (16) FIG. 7 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;
- (17) FIG. 8 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;

(18) FIG. 9 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;

(19) FIG. 10 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;

(20) FIG. 11 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;

(21) FIG. 12 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;

(22) FIG. 13 illustrates a view of an example protection cup for a hearing protection device in accordance with various embodiments of the present disclosure;

(23) FIGS. 14 and 15 are charts illustrating performance of example protection cups for hearing protection devices in accordance with various embodiments of the present disclosure;

(24) FIG. 16 is a schematic diagram illustrating the evaluation area for attenuation performance analysis in accordance with various embodiments of the present disclosure;

(25) FIG. 17 is a chart illustrating insertion loss performance of example protection cups for hearing protection devices in accordance with various embodiments of the present disclosure;

(26) FIGS. 18 and 19 are charts illustrating insertion loss performance of example protection cups for hearing protection devices in accordance with various embodiments of the present disclosure;

(27) FIG. 20 is an example protection cup for a hearing protection device manufactured in accordance with various embodiments of the present disclosure;

(28) FIG. 21 is an example fractal element panel manufactured in accordance with various embodiments of the present disclosure; and

(29) FIG. 22 is an example protection cup manufactured in accordance with various embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(30) Some embodiments of the present disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the disclosure are shown. Indeed, these disclosures may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout.

(31) The phrases “in one embodiment,” “according to one embodiment,” and the like generally mean that the particular feature, structure, or characteristic following the phrase may be included in at least one embodiment of the present disclosure, and may be included in more than one embodiment of the present disclosure (importantly, such phrases do not necessarily refer to the same embodiment).

(32) The word “example” or “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

(33) If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such component or feature may be optionally included in some embodiments, or it may be excluded.

(34) Referring now to FIGS. 1A-1I, various views of an example protection cup **100** for a hearing protection device in accordance with various embodiments of the present disclosure are illustrated herein. In some examples, protection cup **100** refers to the shell of a hearing protection device (such as earmuffs). For example, the protection cup **100**, in an example hearing protection device, may receive and house noise-absorbing materials, such as a cotton or a foam cushion. In some examples, a sealing section (for example, foam cushion) may seal and enclose the protection cup so that it is insulated. An exterior surface of the sealing section may contact the user's ear when the user is wearing the example hearing protection device.

(35) In some examples, protection cup **100** may be in the shape of an oval cup having its shortest diameter in the transverse direction and the longest diameter in the longitudinal direction. In some examples, a protection cup may be made of rigid plastic material, including, for example, acrylonitrile butadiene styrene (ABS). Additionally or alternatively, other materials may be used for the protection cup **100**.

(36) Protection cup **100**, in accordance with various embodiments of the present disclosure, may be implemented in different kinds of hearing protection devices, including passive hearing protection devices, electronic hearing protection devices, active hearing protection devices, consumer electronic headphones, and/or the like.

(37) Specifically, and with reference to example passive hearing protection devices, the material and structure of the protection cup **100** may be used to attenuate noise. In contrast and in example active hearing protection devices, electronic noise cancellation techniques (such as casting an audio signal to cancel out the environmental noise) may be implemented to further increase noise attenuation. For example, noise cancelling circuitry may be included in an example active hearing protection device. By way of further example, a speaker array may be disposed within the protection cup so that a user wearing an example consumer electronic headphone can still hear some external noise, such as important announcement, from the speaker array.

(38) FIG. 1A illustrates the protection cup **100** that includes an outer surface **125**. In some examples, the outer surface **125** may comprise an outer center portion **127** that is sunken from the surrounding area of the outer surface **125**, which

effectively creates a protruding portion **119** on the inner surface **101**, as illustrated in FIG. **1D** and described hereinafter.

(39) The outer surface **125**, as shown in FIG. **1A**, comprises a pair of knobs **129A** and **129B**. In some examples, a headband may be used to connect a pair of protection cups (for example, via the knobs **129A** and **129B**). The headband may be made of elastic material, such that the headband can be securely positioned on a user's head and the protection cups be securely positioned on the user's ears. Additionally or alternatively, the pair of protection cups may be connected using a neckband so that a user can wear the protection cups and a safety hat at the same time.

(40) FIG. **1B** illustrates protection cup **100** that defines an inner surface **101** opposite to outer surface **125**. In various embodiments of the present disclosure, one or more fractal elements (such as the plurality of longitudinal fractal elements **103**, the plurality of transverse fractal elements **105**, and the plurality of additional fractal elements **111** as shown in FIG. **1B**) may be disposed on and protrude or otherwise extend from the inner surface **101** of the protection cup **100**. As used herein, "fractal elements" refers to a structure that is disposed on and protrudes from a surface (such as, for example, a stiffener). In some examples, a fractal element may be made of rigid plastic material, including, for example, acrylonitrile butadiene styrene (ABS). In some examples, other materials may be used to form a fractal element.

(41) As described herein, one or more fractal elements may intersect, be joined, or otherwise be linked to form a pattern of fractal elements that, in some examples, may provide one or more benefits to noise attenuation. In various embodiments of the present disclosure, the plurality of fractal elements may form one or more patterns on the inner surface of the protection cup. Various example patterns are illustrated in FIG. **1B** and FIGS. **5-13**.

(42) Referring now to FIG. **2**, an example measurement diagram of a fractal element is illustrated. The example measurement diagram illustrates example relationships between the thickness of the protection cup **T**, the width of the fractal element **W**, and the corner radius of the fractal element **R**. For example and as is shown in FIG. **2**, the corner radius **R** of the fractal element may be more than 0.25 times the thickness of the protection cup **T**. Further, the width of the fractal element **W** can be calculated based on the following equation:

$$W = k \times T$$

where **T** is the thickness of the protection cup, and **k** is the ratio between the width of the fractal element and the thickness of the protection cup. In some embodiments, the ratio **k** may be in the range of 0.4 to 0.6 (inclusive). In some embodiments, other ratios may be used. By adjusting the values of **T** and **W**, various embodiments of the present disclosure can, for example, adjust and/or otherwise influence noise attenuation levels of example hearing protection devices, and meet requirements for different noise attenuation ratings.

(43) In some examples, fractal elements may be molded from the inner surface **101** of the protection cup **100**. As described in detail further hereinafter, the protection cup **100** may be made of, for example but not limited to, rigid plastic or resin material. A liquid form of the material may be shaped using a rigid frame ("a mold") with hollowed-out blocks. These hollowed-out blocks resemble the shapes of the fractal elements. Example details of an example molding process are described further in connection with FIG. **20**.

(44) Alternatively or additionally, the fractal elements may be affixed or otherwise attached to the inner surface, example details of which are described further in connection with FIGS. **21-22**. Alternatively or additionally, the fractal elements may be etched from the inner surface **101**.

(45) Referring back to FIG. **1B**, the plurality of fractal elements disposed on the inner surface of the protection cup **100** may include, for example, a plurality of first fractal elements, a plurality of second fractal elements, and/or a plurality of additional fractal elements. The plurality of first fractal elements, the plurality of second fractal elements, and/or the plurality of additional fractal elements may protrude from the inner surface of the protection cup. The plurality of first fractal elements, the plurality of second fractal elements, and/or the plurality of additional fractal elements may form a plurality of regions on the inner surface, as described in detail hereinafter.

(46) Specifically, and with reference to the example shown in FIG. **1B**, the plurality of first fractal elements are the plurality of longitudinal fractal elements **103**, and the plurality of second fractal elements are transverse fractal elements **105**. Further, FIG. **1B** illustrates example additional fractal elements as the plurality of additional fractal elements **111**.

(47) Each of the plurality of longitudinal fractal elements **103** is aligned with a longitudinal direction of the protection cup **100**. As shown in FIG. **1B**, the protection cup **100** may comprise three longitudinal fractal elements **103**. Alternatively or additionally, the protection cup **100** may comprise a different number of longitudinal fractal elements, such as two or fewer and/or greater than three.

(48) Each of the plurality of transverse fractal elements **105** is aligned with a transverse direction of the protection cup **100**. As shown in FIG. **1B**, the protection cup **100** may comprise three transverse fractal elements **105**. Alternatively or additionally, the protection cup **100** may comprise a different number of transverse fractal elements, such as two or fewer and/or greater than three.

(49) In the example shown in FIG. **1B**, the plurality of transverse fractal elements **105** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **103**. Together, longitudinal fractal elements **103** and transverse fractal elements **105** may divide the inner surface **101** into a plurality of regions **109**. Each of the plurality of regions **109** may be in the shape of a rectangle. In some examples, one or more of the regions **109** may be of other shapes, such as square, circle and/or the like. The number of the regions **109** may be based on, for example, the size of the cup. As shown in FIG. **1B**, the inner surface **101** is divided into sixteen regions (including partial regions depicted on the top-left corner, top-right corner, bottom-left corner, and bottom-right corner) by the longitudinal fractal elements **103** and transverse fractal elements **105**. Alternatively or additionally, the number of regions may be less than or more than sixteen.

(50) In the example shown in FIG. **1B**, a plurality of additional fractal elements **111** are disposed within at least one of the plurality of regions **109** and protrude from the inner surface **101**. Each of the additional fractal elements **111** forms one or

more patterns, such as a pattern similar to the capital letter “H” in the English alphabet (hence also referred to as H-shaped fractal elements).

(51) In the example shown in FIG. 1B, each of the additional fractal elements **111** may comprise one or more longitudinal segments **113**, transverse segments **115**, and leg segments **117**. Alternatively or additionally, additional fractal elements **111** may take alternative shapes and/or include one or more longitudinal segment **113**, transverse segments **115**, and leg segments **117**. In some examples, the additional fractal elements **111** may take the form of other letters, such as the letters L, K, F, E, and/or the like. Whereas in other examples, the additional fractal elements **111** may take the form of multi-side shapes, such as triangles, squares, pentagons, and/or the like. In yet further examples, the additional fractal elements **111** may take a random or semi-random pattern.

(52) Referring back to the example shown in FIG. 1B, each longitudinal segment **113** is the “horizontal bar” in the middle of the capital letter “H” pattern formed by each of the additional fractal elements **111**. As illustrated in FIG. 1B, the longitudinal segment **113** of each additional fractal elements **111** is parallel to the plurality of longitudinal fractal elements **103** and aligned with the longitudinal direction of the protection cup **100**.

(53) In the example shown in FIG. 1B, two or more of neighboring additional fractal elements **111** may be connected through the longitudinal segment **113** in the longitudinal direction. For example, as shown in FIG. 1B, the longitudinal segment **113A** of fractal element **111A** is connected to the longitudinal segment **113B** of fractal element **111B**. Additionally or alternatively, at least two of neighboring additional fractal elements are not connected through the longitudinal segment. For example, FIG. 5 illustrates an example embodiment where some neighboring additional fractal elements are not connected to each other, details of which are described hereinafter.

(54) The transverse segments **115** are the “vertical bars” in the capital letter “H” pattern formed by each of the additional fractal elements **111**. As illustrated in FIG. 1B, the pair of transverse segments **115** of each additional fractal elements **111** are parallel to the plurality of transverse fractal elements **105** and aligned with a transverse direction of the protection cup **100**.

(55) The leg segments **117** of each of the plurality of additional fractal elements **111** are disposed at end points of the transverse segments **115**. The leg segments **117** are parallel to the plurality of longitudinal fractal elements **103** and aligned with the longitudinal direction of the protection cup **100**. In other words, the leg segments **117** are parallel to the longitudinal segment **113**. As illustrated in FIG. 1B, the leg segments **117** are at a same distance from the corresponding longitudinal segment **113** in the corresponding additional fractal elements **111**. In some embodiments, the leg segments may be at different distances from the corresponding longitudinal segment.

(56) In the example shown in FIG. 1B, the plurality of additional fractal elements **111** have the same size. In other words, the plurality of additional fractal elements **111** may have the same size for the longitudinal segment **113**, the same size for transverse segments **115**, and the same size for leg segments **117**. In some embodiments, one or more of the plurality of additional fractal elements **111** may have different sizes.

(57) In the example shown in FIG. 1B, each of the plurality of additional fractal elements **111** may be disposed on the same position relative to its corresponding region. In some embodiments, the plurality of additional fractal elements **111** may be disposed on different positions relative to their corresponding regions.

(58) In the example shown in FIG. 1B, each of the plurality of regions **109** may include only one of the additional fractal elements **111**. In some embodiments, more than one fractal element may be disposed within each of the plurality of regions **109**.

(59) Referring now to FIGS. 1C-1I, other views of the example protection cup **100** for a hearing protection device are shown. As shown in FIG. 1C and as described above, the plurality of longitudinal fractal elements **103** and the plurality of transverse fractal elements **105** may divide the inner surface **101** of the protection cup **100** into a plurality of regions **109**. As shown in FIG. 1D and as described above, each of the additional fractal elements **111** may comprise a longitudinal segment **113**, a pair of transverse segments **115**, and four leg segments **117**.

(60) FIGS. 1C and 1D illustrate that the inner surface **101** of the protection cup **100** may comprise a protruding portion **119**. In various embodiments of the present disclosure, the plurality of additional fractal elements **111** are disposed on the protruding portion **119** of the inner surface **101** of the protection cup **100**. As described above and further illustrated in the various analyses described below, disposing the plurality of additional fractal elements **111** on the protruding portion **119** of the inner surface **101** influence and/or improves, in some examples, noise attenuation and other performance of the hearing protection device. Example analyses are described in details in connection with FIGS. 14-19.

(61) FIGS. 1C and 1D further illustrate that the inner surface **101** of the protection cup **100** may comprise a curved portion **121** surrounding the protruding portion **119**. In some embodiments, a plurality of ribs **123** are disposed on the curved portion **121** of the inner surface **101** of the protection cup **100**, and may provide structural integrity support for the protection cup **100**.

(62) FIG. 1E to 1I further illustrate additional and/or alternative views of the example protection cup **100**. In particular, FIG. 1E illustrates a back view of the protection cup **100**; FIG. 1F illustrates a front view of the protection cup **100**; FIG. 1G illustrates a top view of the protection cup **100**; FIG. 1H illustrates a bottom view of the protection cup **100**; and FIG. 1I illustrates an isometric view of the protection cup **100**.

(63) Referring now to FIGS. 3A-3B and 4A-4B, cross-sectional views of example protection cups in accordance with various embodiments of the present disclosure are shown. In particular, FIGS. 3A-3B and 4A-4B illustrate example measurements of example protection cups.

(64) Referring now to FIGS. 3A and 3B, cross-sectional views of an example protection cup **300** in accordance with various embodiments of the present disclosure are illustrated.

(65) The example protection cup **300** may comprise a protruding portion **302** surrounded by a curved portion **304**. As shown

in FIG. 3A, a center of the reference plane **306** (defined by the edges of the curved portion **304**) may be at a distance **D1** to an outer edge of the curved portion **304**. In some embodiments, the distance **D1** may be in the range of 40 mm to 70 mm. In some embodiments (and as shown in FIG. 3A), the distance **D1** is 56.86 mm. In some embodiments, other ranges and values may be used for **D1**.

(66) The angle between a reference plane that is orthogonal to the protruding portion **302** and the reference plane **306** may be **D2**. In some embodiments, the angle **D2** may be in the range of 95 degrees to 99 degrees. In some embodiments (and as shown in FIG. 3A), the angle **D2** is 97 degrees. In some embodiments, other ranges and values may be used for **D2**.

(67) A distance between the protruding portion **302** and the reference plane **306** may be **D3**. In some embodiments, the distance **D3** may be in the range of 28 mm to 32 mm. In some embodiments (and as shown in FIG. 3A), the distance **D3** is 30.40 mm. In some embodiments, other ranges and values may be used for **D3**.

(68) A width of the protruding portion **302** may be **D4**. In some embodiments, the distance **D4** may be in the range of 50 mm to 70 mm. In some embodiments, the distance **D4** is preferably 68.88 mm. In some embodiments, other ranges and values may be used for **D4**.

(69) The protrusion depth of the protruding portion **302** may be **D6**. In some embodiments, the distance **D6** may be in the range of 0.5 mm to 1 mm. In some embodiments (and as shown in FIG. 3A), the distance **D6** is 0.75 mm. In some embodiments, other ranges and values may be used for **D6**.

(70) A distance between the protruding portion **302** and reference plane **308** may be **D11**. In some embodiments, the distance **D11** may be in the range of 5 mm to 15 mm. In some embodiments (and as shown in FIG. 3A), the distance **D11** is 10 mm. In some embodiments, other ranges and values may be used for **D11**.

(71) From the cross-sectional view as shown in FIG. 3A, a radius from an imaginary center of the protection cup **300** to the protruding portion **302** may be **D5**. From the cross-sectional view as shown in FIG. 3B, a radius from an imaginary center of the protection cup **300** to the protruding portion **302** may be **D7**. In some embodiments, the value of **D5** may be 260 mm. In some embodiments, the value of **D7** may be 185 mm. In some embodiments, other values for **D2** and/or **D5** may be used.

(72) FIGS. 4A and 4B illustrate cross-sectional views of an example protection cup **400** in accordance with various embodiments of the present disclosure.

(73) Similar to the protection cup **300** illustrated in FIGS. 4A-4B, the protection cup **400** may comprise a protruding portion **401** and a curved portion **403**. The protection cup **400** may also have similar measurements as protection cup **300**, except measurements related to the belly portion **405**, as described below.

(74) As shown in FIG. 4A, the protection cup **400** further comprises a belly portion **405** that is further protruding from the protruding portion **401**. In this regard, **D5** and **D7** of the protection cup **400** as shown in FIGS. 4A-4B may be different from **D5** and **D7** of the protection cup **300** as shown in FIGS. 3A-3B. For example, the **D5** of protection cup **400** may be 170 mm. As another example, the **D7** of the protection cup **400** may be 100 mm. In some embodiments, other values for **D2** and/or **D5** of the protection cup **400** may be used. These differences may change the eigenfrequency and attenuation performance of the protection cups, as described in details hereinafter.

(75) Referring now to FIGS. 5-13, example protection cups with different patterns formed by fractal elements are illustrated. Example performance analyses of these example protection cups are described in connection with FIGS. 14-19.

(76) Referring now to FIG. 5, an example protection cup **500** is shown. The protection cup **500** comprises a plurality of fractal elements disposed on an inner surface **501** of the protection cup **500**. The plurality of fractal elements may include a plurality of first fractal elements (for example, the plurality of longitudinal fractal elements **502**), a plurality of second fractal elements (for example, the plurality of transverse fractal elements **503**), and a plurality of additional fractal elements **505**.

(77) In the example shown in FIG. 5, each of the plurality of longitudinal fractal elements **502** is aligned with a longitudinal direction of the protection cup **500**. Each of the plurality of transverse fractal elements **503** is aligned with a transverse direction of the protection cup **500**. As shown in FIG. 5, the plurality of transverse fractal elements **503** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **502**. Together, longitudinal fractal elements **502** and transverse fractal elements **503** may divide the inner surface **501** into a plurality of regions **504**, similar to the longitudinal fractal elements **103** and transverse fractal elements **105** described above in connection with FIG. 1B.

(78) A plurality of additional fractal elements **505** are disposed on the inner surface **501** of the protection cup **500**, and particularly, within each of the regions **504**. The additional fractal elements **505** are similar to the additional fractal elements **111** described above in connection with FIG. 1B, and form patterns similar to the capital letter "H" in the English alphabet. Comparing the additional fractal elements **505** in FIG. 5 and the additional fractal elements **111** in FIG. 1B, it is noted that each of the additional fractal elements **505** is connected to no more than one other neighboring additional fractal element in the longitudinal direction via the longitudinal segment **506**. For example, the additional fractal element **505A** and the additional fractal element **505B** are connected through their respective longitudinal segment **506A** and **506B**. The additional fractal element **505A** is not connected to the additional fractal element **505C**.

(79) Referring now to FIG. 6, an example protection cup **600** is shown. The protection cup **600** comprises a plurality of fractal elements disposed on an inner surface **601** of the protection cup **600**. The plurality of fractal elements may comprise a plurality of longitudinal fractal elements **602**, a plurality of transverse fractal elements **603**, and a plurality of additional fractal elements **604**, similar to the plurality of longitudinal fractal elements **103**, the plurality of transverse fractal elements **105**, and the plurality of additional fractal elements **111** described above in connection with FIG. 1B.

(80) In the example shown in FIG. 6, the plurality of fractal elements may include one or more corner fractal elements **605**. The corner fractal elements **605** are disposed on the corners of the inner surface **601** of the protection cup **600**. In some examples, each of the corner fractal elements **605** is at forty-five degrees from one of the longitudinal fractal elements **602** and one of the transverse fractal elements **603**. Additionally or alternatively, one or more of the plurality of the corner fractal

elements **605** are at a different degree than forty-five degrees from one of the longitudinal fractal elements **602** and/or one of the transverse fractal elements **603**.

(81) Referring now to FIGS. **7-10**, various example patterns formed by fractal elements are illustrated.

(82) Referring now to FIG. **7**, an example protection cup **700** is shown. The protection cup **700** comprises a plurality of fractal elements disposed on the inner surface **701** of the protection cup **700**. The plurality of fractal elements comprise a plurality of longitudinal fractal elements **702** and a plurality of transverse fractal elements **703**. Each of the plurality of longitudinal fractal elements **702** is aligned with a longitudinal direction of the protection cup **700**. Each of the plurality of transverse fractal elements **703** is aligned with a transverse direction of the protection cup **700**. The plurality of transverse fractal elements **703** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **702**.

(83) Together, the longitudinal fractal elements **702** and the transverse fractal elements **703** may form an “H-tree” structure on the inner surface **701** of the protection cup **700**. The “H-tree” structure may comprise various levels of H-tree patterns formed by the fractal elements. For example, the longitudinal fractal element **702A** and the transverse fractal elements **703A**, **703B** may form a first-level H-tree pattern **704**. As shown, the first-level H-tree pattern **704** resembles the shape of a capital letter “H” in the English alphabet.

(84) Further, a plurality of first-level H-tree patterns may be connected by the longitudinal fractal elements **702** and the transverse fractal elements **703**, and together they form a second-level H-tree pattern, such as the second-level H-tree pattern **705**. As shown in FIG. **7**, the second-level H-tree pattern **705** also resembles the shape of a capital letter “H” in the English alphabet.

(85) In some embodiments, the relationship between the size of the first-level H-tree pattern **704** and the size of the second-level H-tree pattern **705** is as follows:

$$(86) S_1 = \frac{S_2}{k}$$

where S.sub.1 is the size of the first-level H-tree pattern **704**, S.sub.2 is the size of the second-level H-tree pattern **705**, and k is the sizing ratio. In some embodiments, the sizing ratio k equals to the square root of two (i.e. $\sqrt{2}$). For example, the length of the transverse fractal element **703C** may be $\sqrt{2}$ times the length of transverse fractal element **703A**. In some embodiments, the sizing ratio k may have other values, including, for example, golden ratio (1.6180), silver ratio (2.4142), or bronze ratio 3.3002. Similar sizing calculations may also apply to the H-tree patterns as described in connection with FIGS. **8-10**.

(87) Referring back to FIG. **7**, additional levels of H-tree patterns may be formed by the longitudinal fractal elements **702** and the transverse fractal elements **703**. For example, second-level H-tree patterns may be connected by the transverse fractal elements **703D** and **703E** to form a third-level H-tree pattern. The size of the third-level H-tree pattern may be calculated based on the size of the second-level H-tree pattern and the sizing ratio, similar to those described above in connection with the second-level H-tree pattern and the first-level H-tree pattern.

(88) Referring now to FIG. **8**, an example protection cup **800** is shown. The protection cup **800** includes a plurality of fractal elements disposed on the inner surface **801** of the protection cup **800**. The plurality of fractal elements comprise a plurality of longitudinal fractal elements **802** and a plurality of transverse fractal elements **803**. Each of the plurality of longitudinal fractal elements **802** is aligned with a longitudinal direction of the protection cup **800**. Each of the plurality of transverse fractal elements **803** is aligned with a transverse direction of the protection cup **800**. The plurality of transverse fractal elements **803** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **802**.

(89) As shown in FIG. **8**, the plurality of longitudinal fractal elements **802** and the plurality of transverse fractal elements **803** may form a plurality of H-tree patterns at various levels, and at least one H-tree pattern is not connected to any other H-tree pattern at the same level. For example, the plurality of longitudinal fractal elements **802** and the plurality of transverse fractal elements **803** may form a first-level H-tree pattern **804**, a second-level H-tree pattern **805**, and the third-level H-tree pattern **806**. The third-level H-tree pattern **806** is not connected to any other H-tree pattern at the third level (for example, the third-level H-tree pattern **806'**).

(90) Additionally or alternatively, the plurality of longitudinal fractal elements and transverse fractal elements may form one or more H-tree patterns that are not connected to the same or any other level of H-tree patterns. Additionally or alternatively, the plurality of longitudinal fractal elements and transverse fractal elements may form one or more H-tree patterns that are connected to one or more H-tree pattern at a different level (for example, a first-level H-tree pattern may be connected to a separated second-level H-tree via the plurality of longitudinal fractal elements and/or transverse fractal elements).

(91) Referring now to FIG. **9**, an example protection cup **900** is shown. The protection cup **900** comprises a plurality of fractal elements disposed on the inner surface **901** of the protection cup **900**. The plurality of fractal elements comprise a plurality of longitudinal fractal elements **902** and a plurality of transverse fractal elements **903**. Each of the plurality of longitudinal fractal elements **902** is aligned with a longitudinal direction of the protection cup **900**. Each of the plurality of transverse fractal elements **903** is aligned with a transverse direction of the protection cup **900**. The plurality of transverse fractal elements **903** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **902**.

(92) Similar to the longitudinal fractal elements **702** and the transverse fractal elements **703** described above in connection with FIG. **7**, the plurality of longitudinal fractal elements **902** are aligned in a perpendicular relationship to the plurality of transverse fractal elements **903**, and together they may form one or more levels of H-tree patterns.

(93) In comparison with the fractal elements as shown in FIG. **7**, the fractal elements in FIG. **9** further includes a central longitudinal fractal element **904** and a central transverse fractal element **905**. The central longitudinal fractal element **904** is disposed at the central position in the transverse direction on the inner surface **901**, and is parallel to the longitudinal fractal elements **902**. The central transverse fractal element **905** is disposed at the central position in the longitudinal direction on the inner surface **901**, and is parallel to the transverse fractal elements **903**.

(94) Referring now to FIG. 10, an example protection cup **1000** is shown. The protection cup **1000** comprises a plurality of fractal elements disposed on the inner surface **1001** of the protection cup **1000**. The plurality of fractal elements comprise a plurality of longitudinal fractal elements **1002** and a plurality of transverse fractal elements **1003**. Each of the plurality of longitudinal fractal elements **1002** is aligned with a longitudinal direction of the protection cup **1000**. Each of the plurality of transverse fractal elements **1003** is aligned with a transverse direction of the protection cup **1000**. The plurality of transverse fractal elements **1003** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **1002**.

(95) Similar to the spatial relationship between the transverse fractal elements **903** and the longitudinal fractal elements **902** described above in connection with FIG. 9, the plurality of transverse fractal elements **1003** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **1002**, and together they may form one or more levels of H-tree patterns. The plurality of fractal elements in FIG. 10 further comprise a central longitudinal fractal element **1004** and a central transverse fractal element **1005**, similar to the central longitudinal fractal element **904** and the central transverse fractal element **905** described above in connection with FIG. 9.

(96) While the fractal elements as illustrated in FIG. 10 may form more levels of H-tree patterns than the fractal elements as illustrated in FIG. 9, it is noted that the present disclosure is not limited to the fractal elements forming a particular level of H-tree pattern.

(97) Referring now to FIGS. 11-13, various example patterns formed by fractal elements are illustrated.

(98) Referring now to FIG. 11, an example protection cup **1100** is shown. The protection cup **1100** comprises a plurality of fractal elements disposed on an inner surface **1101** of the protection cup **1100**. For example, the plurality of fractal elements may comprise a plurality of longitudinal fractal elements **1102** and a plurality of transverse fractal elements **1103**. Each of the plurality of longitudinal fractal elements **1102** is aligned with a longitudinal direction of the protection cup **1100**. Each of the plurality of transverse fractal elements **1103** is aligned with a transverse direction of the protection cup **1100**. As shown in FIG. 11, the plurality of transverse fractal elements **1103** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **1102**.

(99) The plurality of longitudinal fractal elements **1102** and the plurality of transverse fractal elements **1103** may divide the inner surface **1101** of the protection cup **1100** into a plurality of regions **1104**. In some embodiments, the regions **1104** are in the shape of rectangles. In some examples, one or more regions **1104** may have a shape different from rectangle. In some embodiments, the regions **1104** are of the same size. In some examples, the regions **1104** may be of different sizes.

(100) Referring now to FIG. 12, an example protection cup **1200** is shown. The protection cup **1200** comprises a plurality of fractal elements disposed on an inner surface **1201** of the protection cup **1200**. For example, the plurality of fractal elements may comprise a plurality of longitudinal fractal elements **1202**, a plurality of transverse fractal elements **1203**, and a plurality of diagonal fractal elements **1204**. Each of the plurality of longitudinal fractal elements **1202** is aligned with a longitudinal direction of the protection cup **1200**. Each of the plurality of transverse fractal elements **1203** is aligned with a transverse direction of the protection cup **1200**.

(101) As shown in FIG. 12, the plurality of transverse fractal elements **1203** are aligned in a perpendicular relationship to the plurality of longitudinal fractal elements **1202**. In some examples, each of the plurality of diagonal fractal elements **1204** is at forty-five degrees from one of the longitudinal fractal elements **1202** and one of the transverse fractal elements **1203**. In some examples, one or more of the plurality of diagonal fractal elements **1204** are at a different degree than forty-five degrees from one of the longitudinal fractal elements **1202** and/or one of the transverse fractal elements **1203**.

(102) The plurality of longitudinal fractal elements **1202**, the plurality of transverse fractal elements **1203**, and the plurality of diagonal fractal elements **1204** divide the inner surface **1201** of the protection cup **1200** into a plurality of regions **1205**. In some embodiments, the regions **1205** are in the shape of triangles. In some examples, one or more regions **1205** may have a shape different from triangle. In some embodiments, the regions **1205** are of the same size, and neighboring regions may share an edge. For example, as shown in FIG. 12, the region **1205A** and region **1205B** share an edge that is part of one of the transverse fractal elements **1203**. In some examples, the regions **1205** may be of different sizes.

(103) Referring now to FIG. 13, an example protection cup **1300** is shown. The protection cup **1300** comprises a plurality of fractal elements **1302** disposed on an inner surface **1301** of the protection cup **1300**. As shown in FIG. 13, the plurality of fractal elements **1302** may divide the inner surface **1301** into a plurality of hexagon regions **1303**. In some embodiments, three of the plurality of hexagon regions **1303** meet at a vertex. For example, hexagon region **1303A**, hexagon region **1303B**, and hexagon region **1303C** meet at vertex **1304**. As such, the plurality of fractal elements **1302** are in the pattern of a hexagonal tiling.

(104) In some examples, the plurality of fractal elements **1302** may form the pattern of a trihexagonal tiling. In such examples, the plurality of fractal elements **1302** may divide the inner surface **1301** of the protection cup **1300** into equilateral triangle regions and regular hexagon regions. The equilateral triangle regions and regular hexagon regions are arranged so that each hexagon region is surrounded by triangle regions, and each triangle region is surrounded by hexagon regions.

(105) In various embodiments of the present disclosure, the plurality of fractal elements may form other patterns. As examples, such patterns may include Peano Curve, Hilbert Curve, Moore Curve, Fibonacci Word Fractal, and/or the like. In some embodiments, the pattern may be fractal or cut-through.

(106) While example patterns are described above, it is noted that the present disclosure is not limited to these examples, and the fractal elements may form other patterns without deviating from the scope of the present disclosure. In some embodiments, the pattern may be triangular, rectangular, polygonal, and/or circular. In some embodiments, the pattern may include H-shaped fractal elements having different sizes. In some embodiments, the plurality of fractal elements may form different patterns on the inner surface of the protection cup. For example, the different patterns may be grouped in portions or quadrants, and each portion or quadrant may have a same or different pattern formed by fractal elements.

(107) As described in details hereinafter, various analyses illustrate the effects that the shape and size of the fractal elements may have on the performance of hearing protection device. In particular, eigenfrequency analysis insertion loss analysis are performed on example apparatuses that embody embodiments of the present disclosure. These analyses show that various embodiments of the present disclosure increase and/or manipulate noise attenuation and improve performance of hearing protection device.

(108) FIGS. **14-17** illustrate example eigenfrequency and attenuation analyses, including, for example, insertion loss analysis. In conducting these analyses, four (4) sets of example apparatuses are used. Each set of example apparatuses embodies features from one of the protection cup **300**, the protection cup **400**, the protection cup **700**, and the protection cup **900**. There are twelve (12) example apparatuses within each set (Mode No. 1 to 12), each having a different outer shape. Example apparatuses with the same mode number in different sets have the same outer shape. Further, cushion inside the example apparatuses are removed, and the apparatuses are stiffly fixed on a hard surface.

(109) While these example apparatuses are described, it is noted that the scope of the present disclosure is not limited to these particular example apparatuses. Results of the eigenfrequency analysis and the attenuation analysis indicate that embodiments of the present disclosure improve noise attenuation.

(110) FIGS. **14-15** illustrate the results of eigenfrequency analysis on the performance of example apparatuses embodying the present disclosure. Here, eigenfrequency (or “natural frequency”) refers to the frequency or frequencies at which the hearing protection device is prone to vibrate. In some examples, the results of eigenfrequency analysis as shown may be based on, for example, computer simulations that include computer modeling of example apparatuses described above, and may provide indications on the performance of example apparatuses in real life.

(111) Referring to FIG. **14**, chart **1400** illustrates the example eigenfrequencies of the four (4) sets of example apparatuses described above. In particular, line **1402** shows the eigenfrequencies of the set of example apparatuses that embody features from protection cup **900** as described above. Line **1404** shows the example eigenfrequencies of the set of example apparatuses that embody features from protection cup **700** as described above. Line **1406** shows the example eigenfrequencies of the set of example apparatuses that embody features from protection cup **400** as described above. Line **1408** shows the example eigenfrequencies of the set of example apparatuses that embody features from protection cup **300** as described above.

(112) Referring to FIG. **15**, chart **1500** illustrates the example eigenfrequencies difference ratios of the four sets of example apparatuses described above. In particular, line **1501** shows the example eigenfrequencies difference ratios between the set of example apparatuses that embody features from protection cup **900** and the set of example apparatuses that embody features from protection cup **300**. Line **1503** shows the eigenfrequencies difference ratios between the set of example apparatuses that embody features from protection cup **700** and the set of example apparatuses that embody features from protection cup **300**. Line **1505** shows the eigenfrequencies difference ratios between the set of example apparatuses that embody features from protection cup **400** and the set of example apparatuses that embody features from protection cup **300**.

(113) As can be seen from FIGS. **14-15**, the protection cup **300** and the protection cup **400** result in similar eigenfrequencies of the example hearing protection devices. For example, line **1505** of FIG. **15** (which illustrates the eigenfrequencies difference ratios between protection cup **400** and protection cup **300**) centers around 0%.

(114) In contrast, the protection cup **700** and the protection cup **900** result in different eigenfrequencies of the example hearing protection devices. For example, there is few overlap between line **1501** and line **1503** of FIG. **15**. As described above, the fractal elements of the protection cup **700** and the fractal elements of the protection cup **900** form different patterns. Therefore, FIGS. **14-15** illustrate that fractal elements disposed on the example protection cup can have an impact on the acoustics characteristics of the hearing protection device.

(115) FIGS. **16-17** illustrate the attenuation analysis on the performance of example apparatuses embodying the present disclosure and its result. Referring now to FIG. **16**, a schematic diagram illustrating the model that is used for attenuation analysis is shown. In particular, sound pressure is evaluated over the area **1602**, which has a diameter of ½ inch (i.e. standard measurement microphone). For the attenuation analysis, acoustic attenuation is defined as:

$$(116) (\text{SPL})_{\text{attenu}} = 20 \times \log_{10} \left(\frac{P_{1.\text{sub.amp}}}{P_{0.\text{sub.amp}}} \right) \text{dBSPL}$$

where P0.sub.amp is the pressure amplitude for the model without protection cup, and P1.sub.amp is the pressure amplitude for the model with protection cup.

(117) Referring now to FIG. **17**, results of the attenuation analysis on the set of the example apparatuses are shown. Such attenuation analysis includes, for example, insertion loss (which indicates the loss of noise signal power). In some examples, the results of attenuation analysis as shown may be based on, for example, computer simulations that include computer modeling of example apparatuses described above, and may provide indications on the performance of example apparatuses in real life.

(118) In particular, FIG. **17** illustrates the insertion loss at different frequencies. Line **1701** shows the insertion loss at different frequencies for the set of example apparatuses that embody features from protection cup **300**. Line **1703** shows the insertion loss at different frequencies for the set of example apparatuses that embody features from protection cup **400**. Line **1705** shows the insertion loss at different frequencies for the set of example apparatuses that embody features from protection cup **700**. Line **1707** shows the insertion loss at different frequencies for the set of example apparatuses that embody features from protection cup **900**.

(119) FIG. **17** illustrates the differences in insertion loss of the four sets of example apparatuses described above. In particular, line **1709** illustrates the differences in insertion loss between the set of example apparatuses that embody features from protection cup **400** and the set of example apparatuses that embody features from protection cup **300**. Line **1711**

illustrates the differences in insertion loss between the set of example apparatuses that embody features from protection cup **700** and the set of example apparatuses that embody features from protection cup **300**. Line **1713** illustrates the differences in insertion loss between the set of example apparatuses that embody features from protection cup **900** and the set of example apparatuses that embody features from protection cup **300**.

(120) As shown in FIG. **17**, the insertion loss of the set of example apparatuses that embody features from protection cup **400** become worse (i.e. less noise being reduced) at the frequency range **1715** (as shown by line **1703**), while the insertion loss of the set of example apparatuses that embody features from protection cups **700** and **900** become better at frequency range **1717** (as shown by lines **1705** and **1707**). In some example apparatus embodying features from the protection cup **400**, there is no fractal element. Therefore, FIG. **17** demonstrates that example protection cups in accordance with embodiments of the present disclosure (including protection cup **300**, protection cup **700**, and protection cup **800**) improves noise attenuation, including, for example, insertion loss at least due to the addition of the fractal elements.

(121) FIGS. **18** to **19** illustrate example insertion loss analyses. In conducting these analyses, three apparatuses are used. The first example apparatus is a hearing protection cup without any fractal element, similar to the protection cup **2200** (as shown in FIG. **22**). The second example apparatus embodies features from the example protection cup **800** as shown in FIG. **8**. The third example apparatus embodies features from the example protection cup **1200** as shown in FIG. **12**. In some examples, the results of insertion loss analysis as shown in FIG. **18** may be based on, for example, computer simulations that include computer modeling of example apparatuses described above, and may provide indications on the performance of example apparatuses in real life.

(122) Referring now to FIG. **18**, the chart **1800** illustrates the example attenuation of the example apparatuses described above. In particular, line **1801** illustrates the attenuation performance of the first example apparatus (i.e. without fractal elements). Line **1803** illustrates the attenuation performance of the second example apparatus (i.e. example protection cup **800** as shown in FIG. **8**). Line **1805** illustrates the attenuation performance of the third example apparatus (i.e. example protection cup **1200** as shown in FIG. **12**).

(123) As illustrated the chart **1800**, the attenuation of hearing protection device can be manipulated by the fundamental vibration mode of the hearing protection device via the addition of fractal elements. In particular, lines **1803** and **1805**, in comparison to line **1801**, indicate that fundamental vibration mode of the hearing protection device is adjusted after the fractal elements are implemented on the hearing protection cup. The higher the fundamental vibration mode of the hearing protection device, the higher the attenuation level that the hearing protection device can achieve in the concerned frequency range. In various embodiments, the attenuation level can be manipulated in the concerned frequency range from 500 Hz to 4000 Hz.

(124) Further, by comparing line **1803** and line **1805**, the chart **1800** shows that the specific attenuation level of hearing protection device can be adjusted by varying the shape/dimension of the fractal elements.

(125) Referring now to FIG. **19**, chart **1900** further illustrates the insertion loss analyses of the first example apparatus and the third example apparatus in real life. Here, insertion loss refers to the loss of noise due to hearing protection device. As shown in FIG. **19**, the insertion loss provided by the third example apparatus (i.e. as shown by line **1902**) is superior in comparison to the first example apparatus (i.e. as shown by line **1904**). The affected frequency range is from 500 Hz to 4000 Hz, and the maximum attenuation change can be up to 15 dB.

(126) While the present disclosure is not limited to any particular frequency range, it is noted that the frequency range of 500 Hz to 1500 Hz is of particular interest to the universal rating methods for hearing protection devices, such as Noise Reduction Rating (NRR), SNR (Single Number Rating), and SLC (Sound Level Conversion). Thus, together with considerable attenuation level that can be achieved as shown in FIGS. **18** and **19**, the attenuation of the hearing protection device can be manipulated in a sizable range via, for example, the fractal elements described above to meet the attenuation ratings of the hearing protection device. In other words, various embodiments include a method for manipulating earmuff cup attenuation in a sizable range via the fractal elements to meet the attenuation ratings of the hearing protection device.

(127) Embodiments of the present disclosure may be implemented as methods for manufacturing protection cup and hearing protection device in accordance with various embodiments of the present disclosure.

(128) Referring now to FIG. **20**, an example protection cup **2000** manufactured through a single molding process is shown. The protection cup **2000** may be made of, for example but not limited to, rigid plastic or resin material. A liquid form of the material may be shaped using a rigid frame ("a mold") with hollowed-out blocks. The liquid hardens inside the mold and adopts its shape, resulting the protection cup **2000** as shown in FIG. **20**.

(129) In accordance with various embodiments of the present disclosure, an example method for manufacturing protection cup **2000** may include molding a plurality of longitudinal fractal elements on an inner surface of the protection cup, molding a plurality of transverse fractal elements on the inner surface of the protection cup, and molding a plurality of additional fractal elements on the inner surface of the protection cup.

(130) As described above, the plurality of transverse fractal elements are in perpendicular arrangements to the plurality of longitudinal fractal elements, forming a plurality of regions on the inner surface. Each of the plurality of additional fractal elements are within one of the plurality of the regions on the inner surface.

(131) As described above, each of the plurality of additional fractal elements molded through the example manufacturing process may comprise a longitudinal segment and a pair of transverse segments. In some embodiments, the longitudinal segment is parallel to the plurality of longitudinal fractal elements, and the pair of transverse segments are parallel to the plurality of transverse fractal elements. In some examples, two or more of the plurality of additional fractal elements molded through the example manufacturing process are connected through longitudinal segments of the two or more of the plurality of additional fractal elements.

(132) As described above, each of the plurality of additional fractal elements molded through the example manufacturing process further comprises a plurality of leg segments. Each of the leg segments is molded at an end point of the transverse segments, and the leg segments are molded parallel to the plurality of longitudinal fractal elements.

(133) As described above, the plurality of additional fractal elements molded through the example manufacturing process may be of the same size.

(134) In accordance with various embodiments of the present disclosure, the example method for manufacturing protection cup **2000** also includes forming a protruding portion on the inner surface of the protection cup. In some embodiments, plurality of additional fractal elements are molded on the protruding portion of the inner surface of the protection cup.

(135) In accordance with various embodiments of the present disclosure, the example method for manufacturing protection cup **2000** also includes forming a curved portion on the inner surface surrounding the protruding portion, and molding a plurality of ribs on the curved portion of the inner surface of the protection cup.

(136) Referring now to FIGS. **21-22**, example components for manufacturing a protection cup through an example method in accordance with various embodiments of the present disclosure are shown. The fractal element panel **2100** (as shown in FIG. **21**) and the protection cup **2200** (as shown in FIG. **22**) may be made of, for example but not limited to, rigid plastic or resin material. The fractal element panel **2100** comprises a plurality of longitudinal fractal elements, a plurality of transverse fractal elements, and a plurality of additional fractal elements. As described above, the plurality of transverse fractal elements may be perpendicular to the plurality of longitudinal fractal elements, forming a plurality of regions having a plurality of additional fractal elements. The protection cup may be manufactured by attaching the fractal element panel **2100** with ribs (as shown in FIG. **21**) to the protection cup **2200** (as shown in FIG. **22**).

(137) It is to be understood that the disclosure is not to be limited to the specific embodiments disclosed, and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation, unless described otherwise.

Claims

1. A protection cup for a hearing protection device, comprising: an outer surface; an inner surface on an opposite side from the outer surface; and a plurality of first fractal elements disposed on the inner surface, wherein each of the plurality of first fractal elements comprises a longitudinal segment and at least one transverse segment forming an H-shaped fractal element.
2. The protection cup for the hearing protection device according to claim 1, further comprising a plurality of second fractal elements and a plurality of third fractal elements protruding from the inner surface of the protection cup.
3. The protection cup for the hearing protection device according to claim 2, wherein the plurality of second fractal elements are longitudinal fractal elements and the plurality of third fractal elements are transverse fractal elements, wherein the transverse fractal elements are positioned perpendicular to the longitudinal fractal elements.
4. The protection cup for the hearing protection device according to claim 3, wherein the plurality of second fractal elements and the plurality of third fractal elements form a plurality of regions on the inner surface.
5. The protection cup for the hearing protection device according to claim 1, wherein two or more of the plurality of first fractal elements are connected through longitudinal segments of the two or more of the plurality of first fractal elements.
6. The protection cup for the hearing protection device according to claim 1, wherein each of the plurality of first fractal elements further comprise a plurality of leg segments, wherein the plurality of leg segments is disposed at end points of the at least one transverse segment.
7. The protection cup for the hearing protection device according to claim 1, wherein the plurality of first fractal elements is molded on the inner surface of the protection cup.
8. The protection cup for the hearing protection device according to claim 1, wherein the plurality of first fractal elements is of the same size.
9. The protection cup for the hearing protection device according to claim 1, wherein the inner surface of the protection cup comprises a protruding portion.
10. The protection cup for the hearing protection device according to claim 1, wherein the plurality of first fractal elements further comprises a plurality of corner fractal elements.
11. A method for manufacturing a protection cup for a hearing protection device comprising: molding an outer surface; molding an inner surface on an opposite side from the outer surface; and molding a plurality of first fractal elements on the inner surface, wherein the plurality of first fractal elements comprises a longitudinal segment and at least one transverse segment forming an H-shaped fractal element.
12. The method for manufacturing the protection cup for the hearing protection device according to claim 11 further comprising: molding a plurality of second fractal elements protruding from the inner surface of the protection cup; and molding a plurality of third fractal elements protruding from the inner surface of the protection cup.
13. The method for manufacturing the protection cup for the hearing protection device according to claim 12, wherein the plurality of second fractal elements are longitudinal fractal elements and the plurality of third fractal elements are transverse fractal elements, wherein the transverse fractal elements are positioned perpendicular to the longitudinal fractal elements.
14. The method for manufacturing the protection cup for the hearing protection device according to claim 12, wherein the plurality of second fractal elements and the plurality of third fractal elements form a plurality of regions on the inner surface.
15. The method for manufacturing the protection cup for the hearing protection device according to claim 11, wherein two or

more of the plurality of first fractal elements are connected through longitudinal segments of the two or more of the plurality of first fractal elements.

16. The method for manufacturing the protection cup for the hearing protection device according to claim 11, wherein each of the plurality of first fractal elements further comprises a plurality of leg segments, wherein the plurality of leg segments is molded at end points of the at least one transverse segment.

17. The method for manufacturing the protection cup for the hearing protection device according to claim 11, wherein the plurality of first fractal elements is of the same size.

18. The method for manufacturing the protection cup for the hearing protection device according to claim 11 further comprising forming a protruding portion on the inner surface of the protection cup.

19. The method for manufacturing the protection cup for the hearing protection device according to claim 18 further comprising molding the plurality of first fractal elements on the protruding portion of the inner surface of the protection cup.

20. The method for manufacturing the protection cup for the hearing protection device according to claim 11, wherein molding the plurality of first fractal elements on the inner surface further comprises adjusting the plurality of first fractal elements based on attenuation ratings.
